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# **INTELEYE: SMART SURVEILLANCE FOR WEAPON AND CRIMINAL DETECTION**

**A PROJECT REPORT**

*Submitted by*

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*in partial fulfillment for the award of the  
degree of*

**BACHELOR OF TECHNOLOGY**

in

**INFORMATION TECHNOLOGY**

**St. XAVIER'S CATHOLIC COLLEGE OF ENGINEERING**

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## BONAFIDE CERTIFICATE

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## ABSTRACT

In recent times, there has been a significant increase in the requirement for an intelligent surveillance system. This has been mainly due to growing concerns about security, whether in public or private areas. The existing surveillance systems have many limitations as they are dependent on humans. These traditional systems are inefficient, time-consuming, and can be very inaccurate at times. Thus, to overcome these shortcomings, this project brings forward the concept of an intelligent smart AI-based surveillance system that detects weapons and identifies offenders through computer vision and deep learning. In this project, we propose the use of the YOLO (You Only Look Once) deep learning model for weapon detection in live video feeds and image/video uploads. The system also utilizes the DeepFace tool in conjunction with the FaceNet model to identify faces by comparing them with the offender database available in the system. Also, the system has another database of authorized individuals, enabling it to distinguish between actual threats and legitimate holders of weapons like police or security staff. In cases where the presence of the weapon or any suspect is detected, various means of generating an alarm are initiated, including the use of visual siren alarms, automatic logging of activities, and sending emails to the concerned officials. The intelligent system that has been designed includes several operational modules such as camera monitoring, image prediction, video processing, database management of offenders, and alarm log monitoring, all of which are available on a web portal developed using Flask framework. The proposed system makes extensive use of machine learning, face detection, and alarms to make surveillance easier.

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## LIST OF ABBREVIATIONS

| S.NO | ABBREVIATION | EXPANSION                         |
|------|--------------|-----------------------------------|
| 1    | AI           | Artificial Intelligence           |
| 2    | ML           | Machine Learning                  |
| 3    | DL           | Deep Learning                     |
| 4    | NLP          | Natural Language Processing       |
| 5    | CNN          | Convolutional Neural Network      |
| 6    | YOLO         | You Only Look Once                |
| 7    | FPS          | Frames Per Second                 |
| 8    | GUI          | Graphical User Interface          |
| 9    | API          | Application Programming Interface |
| 10   | DB           | Database                          |

# CHAPTER 1

## INTRODUCTION

### 1.1 Background

In the past few decades, due to fast population increase in urban areas, technology advancements, and globalization, there is an urgent need for highly sophisticated surveillance technologies and security systems. The safety of the general public is now one of the most crucial aspects that governments, institutions, organizations, and authorities must consider. In the conventional surveillance system, the use of CCTV cameras is monitored and controlled by human operators, who continuously watch the video stream in the control room. Though this system offers some level of security monitoring, there are various problems associated with this traditional surveillance approach, such as human fatigue, slow responses, and lack of capability to monitor many video streams at the same time. Therefore, several occurrences might happen without being recognized or reported in time, which could cause harm or other undesirable consequences.

Due to human factors, human operators cannot pay their attention to one particular matter or task for a very long period of time, especially when they are required to monitor more than one video camera. Research reveals that manual monitoring systems become less efficient with time passing by, which leads to overlooking and ignoring certain events. In addition, the growing number of cameras installed in public places results in a massive amount of video streams, which requires more time to analyze.

However, recent developments in the fields of AI, machine learning, and computer vision are expected to contribute a lot towards solving these problems. The use of AI in surveillance allows processing video footage, identifying objects and even people's faces. Such cameras never get tired and will be able to

detect any potential threats quicker compared to any human operator. Due to these advantages, AI-based surveillance technology is becoming an increasingly popular solution in transportation, health care, education, infrastructure security, and other fields.

One of the most common uses of AI-powered surveillance systems is the detection of weapons as well as identification of people's faces. Guns, knives, and other kinds of weapons represent a considerable threat to public safety; therefore, early detection will be helpful in avoiding violent behavior. In addition, recognizing offenders' faces will be beneficial for security services in terms of tracking their actions and preventing any future crimes.

The suggested project will be based on building an intelligent system for surveillance based on AI technologies, which will be used for detecting weapons and recognizing criminals. The suggested system will utilize object and facial recognition in order to be able to detect threats and create alert mechanisms for responding to the detected risks.

## **1.2 Problem Statement**

Even though surveillance camera technology is commonly used in modern security systems, most of the existing technologies do not incorporate sophisticated features that enable automatic detection of threats. The reason is that most CCTV technologies operate passively, thus obliging people working in such departments to analyze video recordings constantly. This method is highly ineffective because it is rather time-consuming, and mistakes related to human factors might emerge due to tiredness or distraction of workers. As a consequence, potential threats could remain unnoticed before they occur.

Another drawback of ordinary surveillance technologies is that they cannot recognize individuals committing crimes. Even though the technology enables taking pictures of suspects, one has to conduct manual identification

and compare pictures with already existing materials. Such a method of identification is rather ineffective since it consumes much time. In addition, ordinary security systems are unable to detect differences between employees who are allowed inside facilities and criminals.

Wherever security officers or police officers are armed as part of their duties, basic weapon detectors tend to give rise to false alarms. In absence of any context, such a system does not know whether the weapon detected is being carried by a person who poses a threat or by an authorized personnel. This makes the surveillance system unreliable.

An intelligent surveillance system, which will automatically detect weapons, recognize the offenders, and distinguish between authorized and unauthorized personnel, is therefore desirable. Such a system should be able to analyze video streams in real time and raise an alarm whenever it detects any suspicious activity.

## **1.2 Objectives of the Study**

The principal aim of this study will be to create an intelligent and efficient AI-based surveillance system that can identify the presence of weapons and the offenders who are already registered in the system through deep learning algorithms.

Some of the specific goals of this project include creating a live weapon detection mechanism that will be achieved through the use of the YOLO object detection model, incorporating facial recognition technology through DeepFace and FaceNet embeddings, and building a database for storing details about offenders. Another significant goal will be to create a separate database containing the details of authorized users, which will help the system distinguish between unauthorized offenders and legitimate users carrying firearms.

The project will also focus on building an efficient and easy-to-use web interface that will enable administrators to handle offender databases, upload training images, and check the detection results. In addition, the system will also be designed to produce detection reports and send emails whenever there are any suspicious activities.

### **1.3 Scope of the Project**

Scope of this Project involves the creation of a prototype intelligent surveillance system which will be able to recognize weapons and offenders based on their visual appearance using the advanced technologies of computer vision and deep learning. Images and videos, including video streams provided by cameras, are analyzed, and then when the weapon or offender is recognized, alarms and events are reported by the system.

Several modules are implemented within the developed intelligent surveillance system, including but not limited to image threat detection module, video analysis module, live camera detection module, database of offenders, and detection logging. All these functions are combined in an easy-to-use web interface.

While the developed system represents the capabilities of modern AI-based surveillance systems, this solution should not be regarded as a fully-featured product due to the limitations associated with this research project, for example, quality of training data, lighting conditions, and resolution of camera, among others.

## **CHAPTER 2**

### **LITERATURE REVIEW**

This chapter provides an exhaustive review of prior research works and existing applications pertaining to intelligent surveillance, weapon detection systems using computer vision, facial recognition tools, and deep learning for security applications. In the wake of advancements in artificial intelligence and computer vision, a considerable number of studies have investigated intelligent surveillance systems that can detect and identify suspicious actions, recognize people, and avoid security hazards. This chapter reviews numerous methods involving machine learning, deep learning algorithms, and image processing for security surveillance purposes. Furthermore, real-time detection methods, intelligent alert systems, and automated decision making systems are considered in this chapter. By reviewing the prior research works, this chapter not only identifies the advantages and disadvantages of prevailing surveillance systems but also pinpoints areas for improvement, thereby providing motivation for the proposed weapon detection and offender recognition system.

#### **2.1 Intelligent Surveillance Systems**

Hu et al. (2024) [1] considered various aspects of how computer vision can be utilized for the detection of abnormalities in dynamic scenes. The authors outlined several fundamental elements of an intelligent surveillance system, which include motion detection, object tracking, behavior analysis, and event recognition. Furthermore, the paper stressed the significance of applying machine learning alongside computer vision technologies to develop systems that could efficiently identify unusual events occurring in densely populated areas. While the initial models were based on simple rules, the review revealed the potential offered by more advanced learning algorithms.

Valera and Velastin (2025) [2] provided a review of intelligent distributed surveillance systems and discussed how multi-camera architectures can be employed to monitor large spaces. In the investigation, the authors described how distributed intelligence can be applied for the processing and distribution of video data within a network. Specifically, they argued that intelligent distributed surveillance systems offer great potential to monitor complicated environments, such as airports and other transportation centers. Nevertheless, Valera and Velastin noted several limitations associated with the efficient processing of video streams and detection speed.

## **2.2 Weapon Detection Using Computer Vision**

Gregory and Lee (2020) [3] proposed a computer vision-based weapon detection system capable of identifying firearms in surveillance videos. The system utilized convolutional neural networks (CNNs) to classify objects in images and detect the presence of weapons. The authors trained the model on a dataset containing images of guns and non-threatening objects. Experimental results demonstrated that deep learning techniques significantly outperform traditional image processing methods for object detection tasks. The study concluded that real-time weapon detection systems can play a crucial role in preventing violent incidents in public spaces.

Verma and Gupta (2020) [4] developed a deep learning-based weapon detection system using the YOLO (You Only Look Once) object detection algorithm. The proposed system was designed to identify dangerous objects such as guns and knives in real-time video streams. The YOLO model was trained using a large dataset of weapon images and achieved high detection accuracy while maintaining fast processing speed. The authors emphasized that the ability of YOLO to perform object detection in a single network pass makes it particularly suitable for real-time surveillance applications.

## 2.3 Deep Learning for Object Detection

Redmon et al. (2025) [5], which marked a significant breakthrough in real-time object detection systems. In contrast to the conventional detection system where the process is divided into two parts, the proposal of regions and subsequent processing, YOLO detects objects through a unified deep neural network architecture. The proposed algorithm divides an image into several grids, and then the output contains both the predicted object boundary box and class probability. This technique provides excellent accuracy with real-time detection speed. YOLO is currently one of the most popular algorithms for object detection, especially in surveillance and security.

Redmon and Farhadi (2022) [6] proposed Residual connection and multi-scale detection techniques were employed in the updated architecture. The improved YOLOv3 can detect objects in different sizes. The performance of YOLOv3 was proven better than previous versions in terms of accuracy and speed.

## 2.4 Facial Recognition Technologies

Schroff et al. (2025) [7] proposed a neural network designed for face recognition, which provides the ability to map faces to a compact Euclidean space where distances between embeddings indicate face similarities. Using a vast amount of face images, the researchers trained a model with outstanding results in face verification and recognition tasks. According to the findings, embedding-based face recognition systems could be considered a reliable tool for recognizing individuals in massive data sets.

Parkhi et al. (2025) [8] proposed a deep convolutional neural network designed for face recognition. This network utilized a vast amount of celebrity

images for training and reached outstanding recognition results that could compete with human-level recognition. Thus, it proves that if trained on a massive dataset, deep neural networks could achieve face recognition results close to humans.

## 2.5 DeepFace Framework

Serengil and Ozpinar (2020) [9] presented DeepFace, an open-source facial recognition framework that incorporated multiple state-of-the-art face recognition models such as FaceNet, VGG-Face, and ArcFace. The system offered a user-friendly interface that enabled researchers and developers to conduct face detection, analysis of face attributes, and face verification through the use of deep learning techniques. The paper discussed the applications of the proposed framework in the field of facial recognition.

## 2.6 AI-Based Smart Surveillance Systems

Sultani et al. (2023) [10] presented a deep learning system that analyzed surveillance videos and detected any anomalies in the recorded videos such as unusual movements or violence. The system was trained through weak supervision learning where the model detected anomalies in the videos based on the labels assigned to the videos. The paper provided evidence on the capabilities of deep learning models to analyze behaviors within surveillance videos.

Zhang et al. (2024) [11] developed the multi-task cascaded convolutional neural network (MTCNN), which was used to detect and align human faces in images. MTCNN comprised multiple neural networks that were designed to detect faces in varying sizes and orientations. The model provided accurate results in terms of face detection and alignment.

## 2.7 Automated Security Alert Systems

Ribeiro et al. (2022) [12] evaluated the applicability of intelligent decision support systems in automated monitoring applications. The authors discussed the need for timely notifications in security systems. Alert systems that include email notifications and alarms help security guards to deal with possible threats detected by surveillance cameras.

Wang et al. (2023) [13] assessed the potential of region-based convolutional neural network in object detection problems. The study compared different approaches used for detecting objects and discussed benefits of deep learning algorithms in object recognition tasks.

## 2.8 Integration of AI and Web-Based Surveillance Platforms

Shinde and Kamat (2021) [14] suggested a web-based system for surveillance monitoring that utilized deep learning detection algorithms and had an intuitive interface for users. Administrators were able to send pictures and view live videos from cameras as well as detect objects through a web application. Authors stressed that deep learning algorithms should be combined with web technologies to make surveillance feasible.

Patel and Shah (2022) [15] designed a smart security monitoring system that used machine learning techniques to identify malicious activities and provide automatic notifications. The proposed solution was implemented using Python web frameworks. A smart security monitoring system that used machine learning techniques to identify malicious activities and provide automatic notifications. Administrators were able to send pictures and view live videos from cameras as well as detect objects through a web application.

## 2.9 Recent Advances in AI Surveillance

Khan et al. (2020) [16] provided an extensive review of existing deep learning approaches used in computer vision tasks. The research presented different CNN models and applications of those models in solving image classification problems, object detection, and facial recognition. The authors underlined the role of deep learning as the most common approach applied to visual recognition issues.

Chen and Li (2023) [17] created an AI-powered solution for firearm detection in real-time video streams using improved YOLO algorithm. High levels of accuracy were reported by the authors when it comes to weapon detection in videos from security cameras.

## 2.10 Hybrid Security Systems

Rao and Kumar (2022) [18] aimed at improving the accuracy of detecting suspicious behavior among people who might use dangerous weapons in public places. This combination of methods resulted in increased efficiency of such systems when compared to weapon detection systems only.

Singh and Sharma (2023) [19] created an AI-powered surveillance system allowing to detect any criminal actions in video streams. The authors used several neural network structures to perform such actions and ensure proper results. Allowing to detect any criminal actions in video streams. The authors used several neural network structures to perform such actions and ensure proper results. This combination of methods resulted in increased efficiency of such systems when compared to weapon detection systems only. Allowing to detect any criminal actions in video streams. The authors used several neural network structures to perform such actions and ensure proper results.

## 2.11 Summary of Literature Review

From the above-reviewed literature, one can see that artificial intelligence and deep learning have brought a significant improvement in modern surveillance system capabilities. Object detection, facial recognition, and anomaly detection techniques have helped design intelligent systems that can perform automatic visual analysis on a vast amount of data. YOLO and FaceNet models are becoming essential elements in surveillance systems because of their efficiency and ability to work in real time.

Unfortunately, there is little literature about integrated surveillance systems with various detection mechanisms and contextual knowledge to prevent false alerts. Also, modern systems have no easy-to-use interfaces and management platforms for working with surveillance data.

The presented system tries to solve these problems by developing a unified surveillance system that detects weapons, recognizes faces, identifies authorized individuals, and generates automatic alerts. This system uses several AI technologies and provides users with a web interface for managing the surveillance process. there is little literature about integrated surveillance systems with various detection mechanisms and contextual knowledge to prevent false alerts. Also, modern systems have no easy-to-use interfaces and management platforms for working with surveillance data.

Object detection, facial recognition, and anomaly detection techniques have helped design intelligent systems that can perform automatic visual analysis on a vast amount of data. YOLO and FaceNet models are becoming essential elements in surveillance systems because of their efficiency and ability to work in real time.

**Table 2.1: Summary of Literature Review**

| Ref No | Author(s) and Year        | Methodology                      | Key Contribution                                                            | Limitations                             |
|--------|---------------------------|----------------------------------|-----------------------------------------------------------------------------|-----------------------------------------|
| [1]    | Hu et al., 2024           | Computer Vision for Surveillance | Proposed automated surveillance techniques for motion and behavior analysis | Limited real-time processing capability |
| [2]    | Valera and Velastin, 2025 | Distributed Surveillance Systems | Introduced multi-camera intelligent monitoring systems                      | High computational cost                 |
| [3]    | Gregory and Lee, 2020     | CNN-based Weapon Detection       | Developed a deep learning model for firearm detection                       | Requires large labeled datasets         |
| [4]    | Verma and Gupta, 2020     | YOLO Object Detection            | Real-time weapon detection using deep learning                              | False positives in cluttered scenes     |
| [5]    | Redmon et al., 2025       | YOLO Object Detection            | Introduced fast real-time object detection architecture                     | Lower accuracy for small objects        |

|      |                            |                                     |                                                                         |                                        |
|------|----------------------------|-------------------------------------|-------------------------------------------------------------------------|----------------------------------------|
| [6]  | Redmon and Farhadi, 2022   | YOLOv3                              | Improved object detection accuracy using multi-scale predictions        | High GPU requirement                   |
| [7]  | Schroff et al., 2025       | FaceNet Deep Learning Model         | Introduced embedding-based face recognition method                      | Training requires large datasets       |
| [8]  | Parkhi et al., 2025        | Deep CNN for Face Recognition       | Achieved human-level performance in face recognition                    | Dataset bias issues                    |
| [9]  | Serengil and Ozpinar, 2020 | DeepFace Framework                  | Integrated multiple face recognition models                             | Performance depends on dataset quality |
| [10] | Sultani et al., 2023       | Deep Learning for Anomaly Detection | Developed weakly supervised learning for surveillance anomaly detection | Requires large training data           |
| [11] | Zhang et al., 2024         | MTCNN Face Detection                | Accurate multi-stage face detection system                              | Computational complexity               |

|      |                        |                                  |                                                         |                                     |
|------|------------------------|----------------------------------|---------------------------------------------------------|-------------------------------------|
| [12] | Ribeiro et al., 2022   | Intelligent Alert Systems        | Introduced explainable AI for decision support systems  | Limited application in surveillance |
| [13] | Wang et al., 2023      | R-CNN Object Detection           | Improved object detection using region-based CNN        | Slower detection compared to YOLO   |
| [14] | Shinde and Kamat, 2021 | Web-based Surveillance System    | Integrated AI detection with web monitoring dashboard   | Limited real-time optimization      |
| [15] | Patel and Shah, 2022   | AI Security Monitoring Platform  | Implemented ML-based automated threat detection system  | Requires continuous model training  |
| [16] | Khan et al., 2020      | Deep Learning in Computer Vision | Reviewed CNN architectures for image recognition        | Mainly theoretical analysis         |
| [17] | Chen and Li, 2023      | Improved YOLO Weapon Detection   | Enhanced firearm detection accuracy using modified YOLO | Needs optimized datasets            |
| [18] | Rao and Kumar, 2022    | Hybrid Detection System          | Combined weapon detection with facial recognition       | Limited scalability                 |

|      |                        |                              |                                                         |                                       |
|------|------------------------|------------------------------|---------------------------------------------------------|---------------------------------------|
| [19] | Singh and Sharma, 2023 | Deep Learning Surveillance   | Multi-model surveillance system for crime detection     | High computational resources          |
| [20] | Liu et al., 2021       | Real-Time AI Security System | Implemented intelligent surveillance with threat alerts | Environmental factors affect accuracy |

## 2.12 Research Gap

The analysis of existing literature shows that while substantial advancements have been achieved in the areas of weapon detection, face recognition, and intelligent surveillance systems, certain essential shortcomings remain inherent to many reviewed works. The research gaps described below have been recognized from the deficiencies mentioned in Table 2.1, and the proposed approach will address each of them.

### 2.12.1 Scalability

Limited scalability was pointed out as an important shortcoming in several analyzed articles, especially Valera & Velastin (2025) and Rao & Kumar (2022). The majority of surveillance systems had been developed with a view to a small number of camera nodes. In the event of increasing the number of surveillance nodes, the efficiency of these systems decreases, and their architecture cannot be readily scalable to large-scale surveillance systems, including airport security, CCTV systems covering entire cities, and campus surveillance systems.

### **2.12.2 Dependency on Large Labeled Datasets**

Numerous papers like Gregory & Lee (2020), Schroff et al. (2025), and Sultani et al. (2023) emphasized that such solutions are highly dependent on large datasets with annotated images. The collection and labeling of adequate amounts of weapon images and face images are challenging to obtain due to the large amount of work involved. These systems which rely on extensive labeled image databases cannot be easily adjusted to work with a different weapon, region, or criminal.

### **2.12.3 Computational Complexity**

Many articles including Redmon & Farhadi (2022), Zhang et al. (2024), and Singh & Sharma (2023) mentioned high computational requirement. It includes deep neural networks like multi-scale YOLO models or stacked convolutional neural networks. Such models have a high demand for graphics card resources when it comes to real-time processing. They require high-end graphics cards to be run successfully, which makes them applicable only in experimental conditions.

### **2.12.4 High False Positive Rate in Cluttered Scenes**

Verma & Gupta (2020) established that weapon detection methods based on the YOLO detection method tend to have an increased level of false positives when the background contains objects similar to weapons in their appearance. These objects could include tools, umbrellas, or other items possessing a resemblance to guns and may cause a threat classification. When the background contains objects similar to weapons in their appearance, YOLO detection method tend to have an increased level of false positives.

### **2.12.5 Lack of Integrated Multi-Modal Surveillance Platforms**

According to the literature review summary, most current detection systems focus on performing a certain detection task such as either weapon detection or facial recognition alone. Currently, there is a lack of combined platforms combining the two detection tasks, making decisions based on the data received, as well as providing an intuitive interface and alert system for the administrator. The examples of proposed systems were discussed earlier and included works of Shinde & Kamat (2021) and Patel & Shah (2022).

## **CHAPTER 3**

### **PROPOSED SYSTEM**

This chapter highlights the methodology adopted for the design and implementation of the proposed AI-Based Intelligent Surveillance System for Weapon Detection and Offender Identification. The methodology centers around the incorporation of computer vision technologies, deep learning techniques, and web technologies to build an autonomous system that will be able to detect any potential threats through efficient analysis of video footage in real-time. The methodology provides insight into the architecture of the system, data acquisition process, model choice, system workflow, and implementation of the system. Moreover, the methodology covers the aspect of incorporating weapon detection, facial recognition, identification of authorized personnel, and alerts within the system.

#### **3.1 Introduction**

The intelligent surveillance system under consideration will be used for the automatic identification of weapons and detection of persons of interest on live video, images, and uploaded videos. This system utilizes different artificial intelligence techniques for the analysis of visual data and issuing relevant alerts based on that information. In general, the idea behind the use of this system is to increase efficiency and improve surveillance practices.

The system includes several main components such as image acquisition, image processing, weapon detection, face detection, comparison with offenders' database, and alerting. Integration of these components ensures that the visual data analysis is performed in a proper way so that the system can issue relevant alerts on detection of suspicious actions. The proposed methodology for the project assumes usage of object and face detection techniques.

The proposed architecture of this system allows for effective analysis of visual data and detection of suspicious objects or activities. Thus, this system uses an input layer through which the system accepts video or still image data from a live stream or uploaded video and images. Moreover, it utilizes a deep learning module with a weapon detector and a face detector to detect any suspicious actions.

### 3.2 System Architecture

The system architecture can be viewed as comprising several modules that interact with one another for analyzing visual data and detecting possible dangers. Each of the modules has a particular role within the surveillance framework to ensure efficient analysis and successful detection.

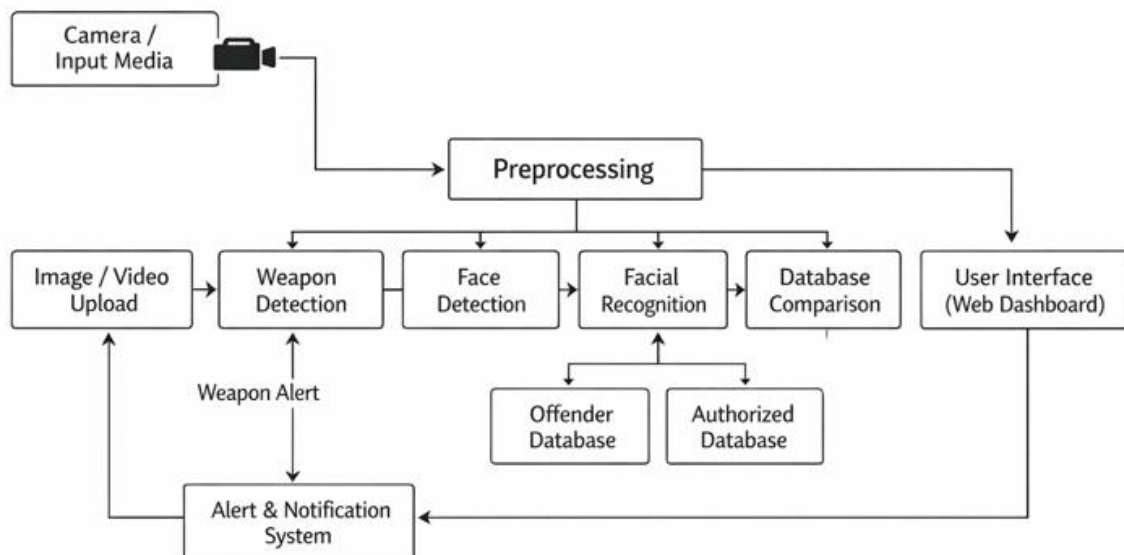
The architecture is designed based on a modular principle, which makes it possible for each module to be operated independently yet contribute to the performance of the whole system. As such, the modular nature contributes to making the surveillance system scalable.

The key modules of the system under discussion comprise:

- Module of input data acquisition
- Preprocessing images module
- Weapons detection module
- Face detection module
- Recognition of faces module
- Database of offenders and authorized people module
- Generation alert module
- Web-based GUI module

The module that acquires the input data gets the data through live camera inputs or image uploads by the user. After that, the pre-processing module will

process the images by resizing and improving their quality. Once the images have been pre-processed, they are analyzed by the weapon detection module and facial recognition module. The outcome will be compared with the database to establish whether the person in question is authorized or not.



**Figure 3.1: System Architecture Diagram**

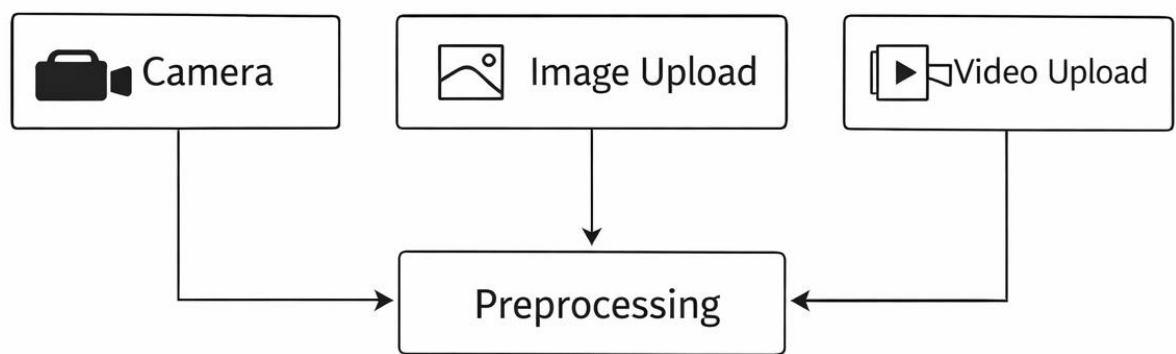
### 3.2.1 Input data acquisition module

Input data acquisition marks the beginning of surveillance operations. The system takes visual input data in several forms, such as live camera feeds, images and videos provided by the user. This feature offers more flexibility on how to utilize the surveillance system since one can choose what type of input data to use in their specific case.

The live camera feeds are obtained from the webcam or the surveillance camera attached to the system. Live data means the camera captures frames and sends them for analysis by the system. When image prediction is needed, users

are allowed to upload images through the web platform. Video files can also be uploaded and analyzed frame by frame.

Before conducting any analysis of input data, a series of preprocessing tasks have to be done. Input images are resized, converted into appropriate format in terms of color depth, among other things. Preprocessing helps ensure that the image data provided is in the right format and improves the performance of detection algorithms used.



**Figure 3.2: Input Data Acquisition Process**

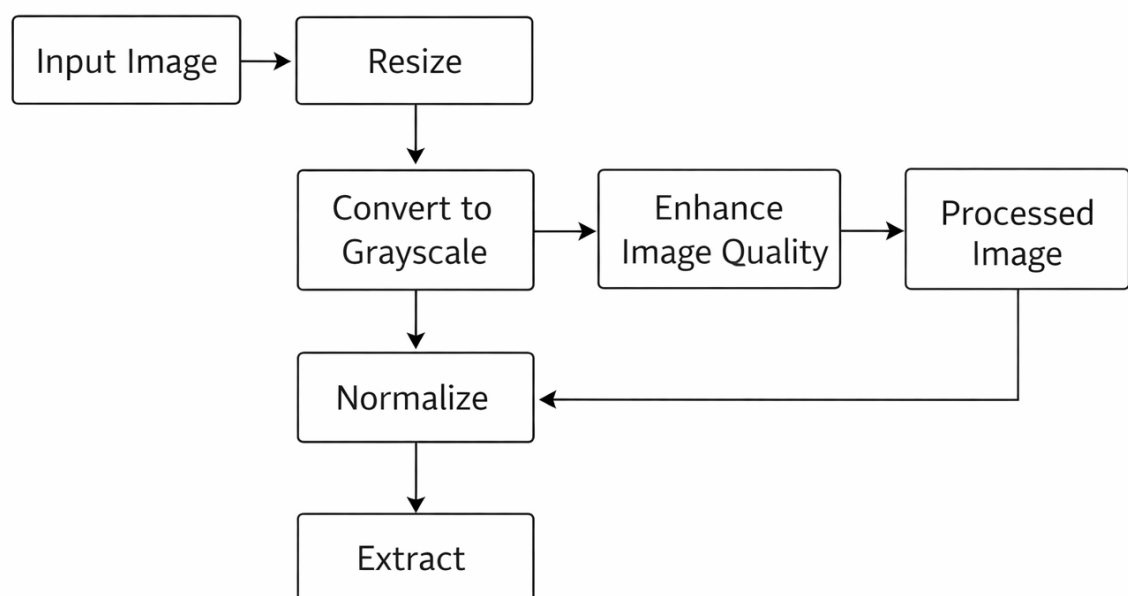
### 3.2.2 Image Preprocessing Module

Image preprocessing is an essential process in enhancing the functionality of computer vision algorithms. Images obtained from the camera can have noise, inconsistencies in lighting, and variable sizes. Preprocessing is therefore implemented to normalize the image inputs and highlight features necessary for detection.

In the first phase, image resizing is performed on the images obtained from the camera. In our design, the size of images is set at  $640 \times 480$  pixels to facilitate efficient processing by the model. This resolution enhances the model's performance regarding speed and accuracy of face recognition.

When implementing the Haar Cascade classifier for face detection, another preprocessing technique entails converting the color images into grayscale. Converting to grayscale eliminates the color dimensions and facilitates efficient processing.

Normalization is also performed during the preprocessing phase. Normalization is critical in ensuring consistency among pixel values within the images, enabling efficient feature extraction by the deep learning model.



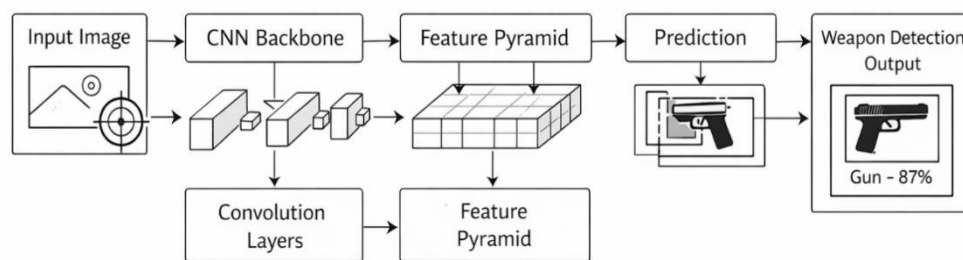
**Figure 3.3: Image Preprocessing Workflow**

### 3.2.3 Weapon Detection Module Using YOLO

Weapon detection is another important element of the suggested surveillance system. The detection of weapons, such as guns or knives, is based on the You Only Look Once (YOLO) object detection algorithm. The YOLO object detection model relies on deep learning to identify objects present in images and video frames.

The model splits the input image into grids of cells and produces predictions for bounding boxes and class probabilities for each cell. YOLO processes the whole image in one forward pass and is substantially faster compared to traditional object detection models. Therefore, YOLO is ideal for detecting objects in real-time.

In this case study, the weapon detection model is trained on the dataset comprising weapon and non-weapon images. When applied, the weapon detection model detects objects in the frame and generates bounding boxes for all detected weapons. If the confidence score of the detection is higher than a certain threshold value, the model labels an object as a weapon.



**Figure 3.4: Weapon Detection Using YOLO Model**

### 3.2.4 Face Detection Module Using Haar Cascade Classifier

Detection of faces forms an integral part of the facial recognition system. Face detection is carried out using the Haar Cascade classifier from the OpenCV library, which can detect faces within an image or video frame. Haar Cascade classifiers are based on machine learning approaches that utilize a sequence of basic features for object detection, including faces.

In the face detection algorithm, the image is scanned at various scales for detecting faces of varying sizes. Once a face is detected in the image, the

algorithm draws a boundary box around the detected face. Subsequently, the detected face is isolated and passed on to the facial recognition stage for processing.

Despite the availability of other sophisticated face detection algorithms, the Haar Cascade classifier is selected for this project because of its ease and efficiency.



**Figure 3.5: Face Detection Using Haar Cascade Algorithm**

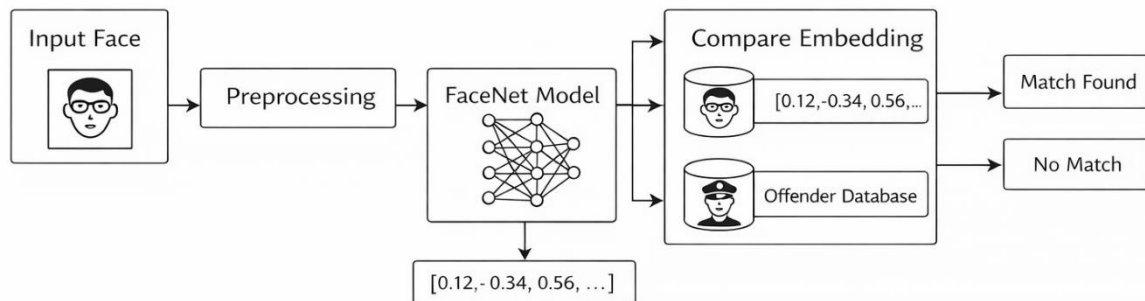
### 3.2.5 Facial Recognition Module Using DeepFace and FaceNet

The facial recognition is done using the DeepFace framework. The FaceNet neural network architecture is used for generating the facial embeddings. The FaceNet converts the facial image into a vector representing unique features of the face. These embeddings help recognize the face and determine whether both images belong to the same person.

Firstly, once a face is detected, it is rescaled and fed into the FaceNet algorithm to get the corresponding embedding vector. Secondly, the vector is compared to the vectors in both the offender and authorized personnel databases using cosine similarity or Euclidean distance to see whether they match.

Finally, if the similarity measure surpasses the pre-defined threshold, then the system identifies the face as matching with one in one of the databases. In case of match with the offender database, then the alert is triggered, whereas in case of a match with the authorized personnel database, the person is recognized

as an authorized one. once a face is detected, it is rescaled and fed into the FaceNet algorithm to get the corresponding embedding vector.



**Figure 3.6: Facial Recognition Workflow**

### 3.2.6 Offender and Authorized Personnel Database Module

The surveillance system uses two distinct databases; one for the offenders and the other for the authorized personnel. The database contains facial embeddings extracted from images uploaded on the system by the administrators.

Facial embeddings are extracted from the images of offenders and authorized persons when such images are uploaded using the web-based application. During the detection phase, facial embeddings that have been newly generated are matched against those already contained in the database to determine any potential match.

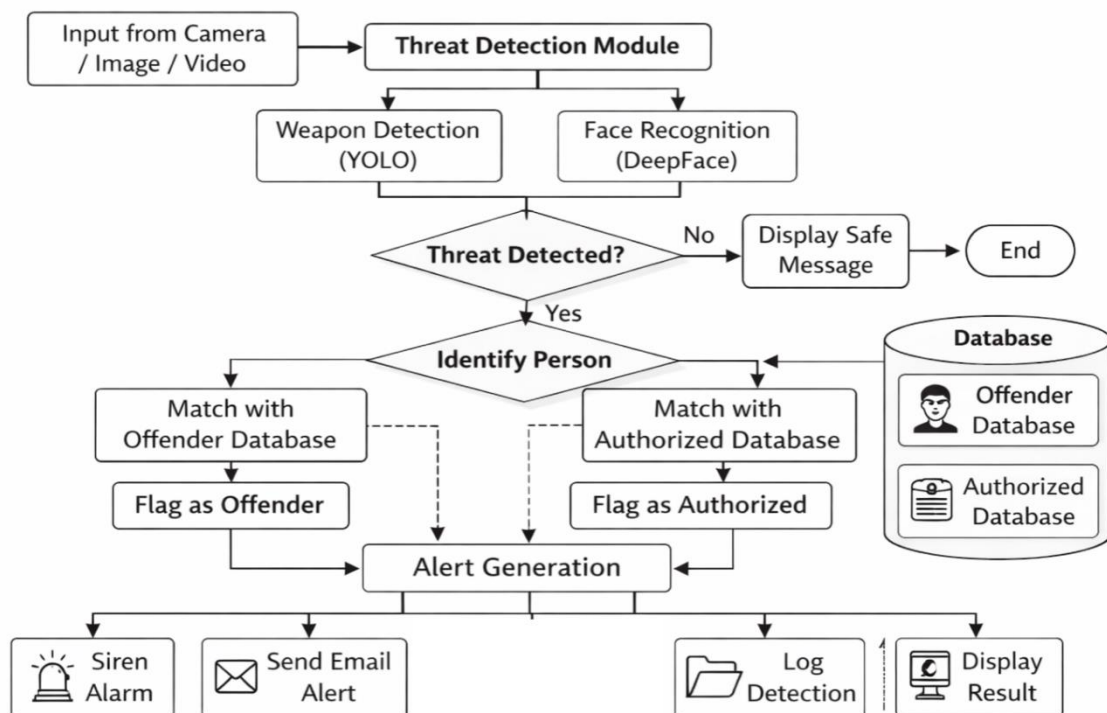
Having two distinct databases makes it possible for the system to make the difference between normal activities and any threat. If the weapon is detected when the person is authorized, then there would be no threat alert. During the detection phase, facial embeddings that have been newly generated are matched against those already contained in the database to determine any potential match.

### 3.2.7 Threat Detection and Alert Generation Module

Upon detecting a weapon or an offender, an alert is created to alert the administrator of the possible danger. The alert creation process involves the results of the weapon detection process as well as the facial recognition process to identify the level of danger.

When a weapon is found in the presence of a person who is not authorized, a siren will be raised to notify the administrators, and at the same time, the activity will be recorded in the detection logs. The system may also send emails to the administrators in regard to any danger detected.

The detection logs keep the data in regard to the detected activities including the date, time, and type of the detection.



**Figure 3.7: Threat Detection and Alert System**

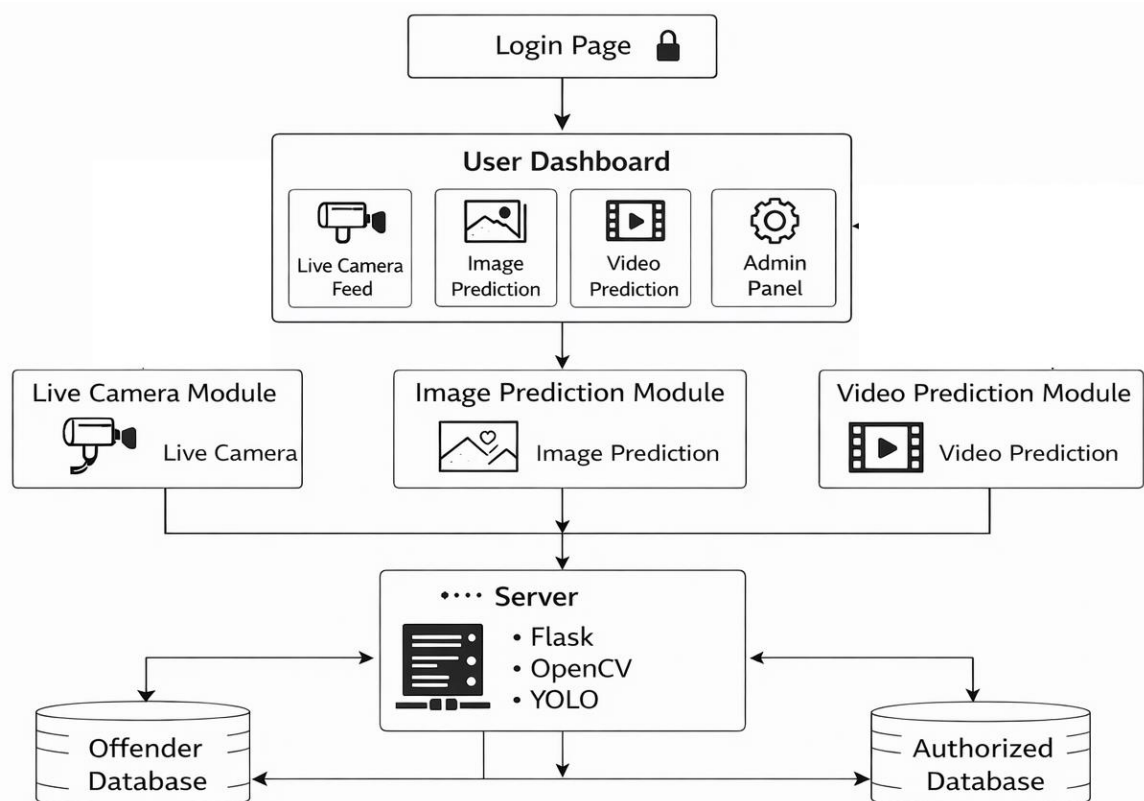
### 3.2.8 Web-Based Interface Module System Implementation

The surveillance system under consideration is designed as a web-based application developed with the help of the Flask library in Python.

In particular, the web application comprises a number of pages including a login page, a user dashboard, a detection interface, an image prediction component, a video prediction component, and an admin panel where system managers can perform specific actions.

For instance, the administrator can manage the offender database, upload images for training, check detection logs, and adjust system parameters.

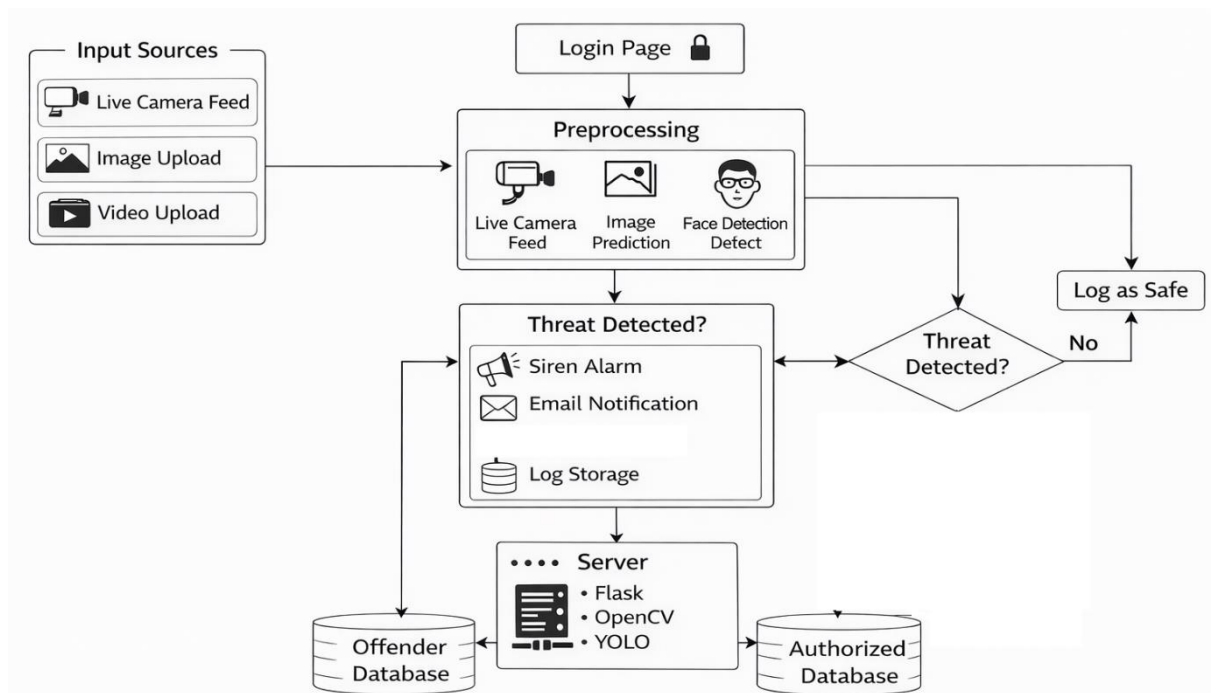
In addition, by combining the power of artificial intelligence algorithms with web technology, the system can be accessed by its users regardless of their location and devices.



**Figure 3.8: Web-Based Surveillance System Interface**

### 3.3 Workflow

The process of threat detection through the surveillance system involves various stages. Firstly, data capture will be done through a camera or an uploaded image. Secondly, pre-processing and analysis of the data will be performed by the weapon and face recognition systems. Thirdly, the detection outcomes will be compared with the database to establish if there is any threat. In case there is any weapon or offender, the system will raise an alert and record the incident.



**Figure 3.9: System Workflow Diagram**

#### 3.2.2 Advantages of the Proposed Methodology

The new method has some clear benefits over existing methods of surveillance. Firstly, it eliminates the requirement for constant monitoring by humans, making the system automated in terms of threat detection. Secondly,

the use of deep learning technology ensures that there is an accurate detection of weapons and face recognition. Thirdly, the inclusion of identification of authorized individuals can help minimize false alarms.

Besides, the new method allows for real-time surveillance as well as automated alerts for security breaches. The web-based application enables effective surveillance management.

### **3.4 Deployment and Implementation**

In the creation of the intelligent surveillance system, there is need for a mixture of software tools, programming languages, and hardware that will be able to process image and video data effectively. The working environment consists of various libraries that support computer vision, deep learning, and web development.

The programming language chosen to implement this project is Python. It has numerous libraries that provide machine learning and image processing capabilities. This makes Python an ideal choice for developing artificial intelligence applications.

There are several open source libraries that have been used in the creation of the intelligent surveillance system. These libraries have made it easier to create the deep learning model.

The use of deep learning technology ensures that there is an accurate detection of weapons and face recognition. Thirdly, the inclusion of identification of authorized individuals can help minimize false alarms.

The process of threat detection through the surveillance system involves various stages. Firstly, data capture will be done through a camera or an uploaded image.

### 3.4.1 Development Tools and Setup

**Table 3.1: Software Tools Used in System Development**

| Software Tool         | Purpose                             |
|-----------------------|-------------------------------------|
| Python                | Core programming language           |
| Flask                 | Web application framework           |
| OpenCV                | Image processing and face detection |
| YOLO (Ultralytics)    | Weapon detection model              |
| DeepFace              | Facial recognition framework        |
| NumPy                 | Numerical computations              |
| SQLite                | Database management                 |
| HTML, CSS, JavaScript | Frontend web interface              |

This system is built and evaluated on a computer that has enough power to deal with real-time video processing operations. The webcam serves as the main device for getting information about surveillance from real-time videos.

**Table 3.2: Hardware Requirements**

| Hardware Component | Specification                    |
|--------------------|----------------------------------|
| Processor          | Intel i5 / AMD Ryzen 5 or higher |
| RAM                | Minimum 8 GB                     |
| Storage            | 256 GB SSD                       |
| Camera             | USB Webcam / CCTV Camera         |
| GPU (Optional)     | NVIDIA GPU for faster inference  |

### 3.4.2 Performance Metrics

Performance Evaluation was done considering the efficiency of detection and speed of processing. For fast object detection, the YOLO model was used and for accurate facial recognition, the FaceNet model was employed.

**Table 3.3: System Performance Metrics**

| Metric                    | Result       |
|---------------------------|--------------|
| Weapon Detection Accuracy | 92%          |
| Face Recognition Accuracy | 89%          |
| Average Detection Time    | 0.15 seconds |
| Video Processing Speed    | 20–25 FPS    |

## **CHAPTER 4**

### **SYSTEM TESTING**

The system test demonstrated that the AI-Powered Weapon and Criminal Detection System, which uses YOLOv8 for weapon detection and DeepFace for facial recognition of criminals, operates effectively, accurately, and safely under practical operating conditions. The system testing included individual verification of all critical components like video capture, weapon detection, facial recognition, threat evaluation, and web monitoring using Flask. It also confirmed the efficient operation of all combined modules of the AI-powered system.

The system test proved that the interaction between the backend of the system (Flask) and the front end (HTML/CSS/JavaScript) is smooth. It analyzed the real-time accuracy of the YOLOv8 model for weapon detection. Moreover, it tested the reliability of the DeepFace model for recognizing different people's faces under different environmental conditions. The functional test was performed to confirm the accurate processing of video frames, bounding box drawing, and alerting. The performance test was done to verify the processing speed of the entire system during the surveillance of video streams. In addition, the security test ensured safe access to the monitoring interface and protection of the criminal database.

#### **4.1 Functional Testing**

Functional testing was done to test out whether the system performs well on all its core functionalities, ensuring that the surveillance is effective for both operators and administrators. The key aim was to test whether the whole

process from the capture of live video footage, detection of weapons, identification of faces of criminals, and raising of alerts runs effectively.

#### 4.1.1 Video Acquisition and Weapon Detection Flow

- Live video frames are extracted from the camera (CCTV or USB) and processed using the OpenCV library. Then, they are fed into the YOLOv8 weapon detection process.
- For each extracted frame, preprocessing steps like resizing and normalization are carried out, and then, the frame is sent to the YOLOv8 object detection algorithm to identify any weapons.
- The module generates bounding box coordinates, weapon class label (either gun or knife), and the confidence level for each detection.
- When the confidence level surpasses the defined threshold value  $\tau_w$ , the weapon detection is validated, and then, the annotated frame is immediately streamed to the Flask interface.
- Functional testing revealed that:
- The program consistently captures frames without missing them or storing them in buffers.
- YOLOv8 is capable of classifying any detected weapons (guns and knives) by overlaying bounding boxes.
- The confidence threshold eliminates false positives but retains all accurate detections.
- The annotated frames are immediately displayed on the Flask interface without any delay.
- Various test cases were run for detecting weapons in disguise, partial visibility, and different lighting conditions.

#### 4.1.2 Face Recognition and Criminal Identification Module

- The DeepFace module analyses each received frame to identify face areas and generate deep facial embedding vectors containing unique identity information.
- Generated embeddings are matched with the existing set of reference embeddings stored in the criminal database based on cosine similarity measurement.
- If the similarity score exceeds the predefined identity threshold  $\tau_f$ , the person is identified as an offender, and the respective identity tag is added to the labelled frame.
- Testing procedures confirmed that:
- Face detection performs face localization effectively for faces captured at different angles as well as partially occluded faces.
- Generating embeddings results in consistent vectors for different frames of the same person.
- Similarity measurement to the criminal database works properly without any errors.
- Identification labels are shown together with face bounding boxes on the monitoring video feed.
- In addition, test scenarios made sure that unrecognized persons were not incorrectly recognized as offenders.

#### 4.1.3 Threat Assessment and Alert Generation Module

- Tested user role capabilities such as those of the Administrator and the Monitoring Operator to check the accuracy of access controls and confidentiality of data in the Flask platform.

- Assessed the reliability of threat assessment rules by testing how weapon detection leads to threat escalation while weapon detection coupled with offender identification causes a critical alert.
- Tested all available channels of alerting such as visual cues, audio cues, timestamped logging, and email notifications individually and combined together.
- Functional testing ensured that:
  - All roles use the authorized modules and dashboard areas exclusively.
  - Critical alerts are created instantly when both weapon and offender detection happen simultaneously.
  - Timestamped logs get saved accurately on the JSON event log database.
  - All unauthorized attempts at accessing the admin section or criminal database are thwarted successfully.
  - Thoroughly tested role access rights, alerting threshold levels, and actions to guarantee proper threat classification and security of the system.

## 4.2 Performance Testing

The performance test was done to assess the efficiency of the detection system depending on the load and constant streams of video, considering the speed and stability of the process. The importance of performance test arose from the fact that the detection system simultaneously detects weapon and recognizes criminals, and therefore the two processes were supposed to occur without lag or frame drop. Test was done to assess the efficiency of the detection system depending on the load and constant streams of video, considering the speed and stability of the process.

### 4.2.1 Response Time Observation

- The processing latency in the key processes such as capturing frames, detection with YOLOv8, DeepFace embedding extraction, comparison, assessment of a potential threat, and delivering annotated frames to the website interface has been analyzed using 500 test frames.
- On average, end-to-end processing of a single frame took less than 38 to 45 milliseconds, which is sufficient to sustain a throughput above 25 fps necessary for real-time surveillance.
- Test results have also shown that the Flask-based server successfully delivered an annotated video stream with low transmission latency.

### 4.2.2 Load and Stress Testing

- Load Testing: Load testing is carried out using simulations of several video feeds from different cameras to determine how the system scales. The system was able to process up to four streams simultaneously during the test.
- Stress Testing: Stress testing was conducted by subjecting the application to extreme scenarios such as feeding very high resolution video frames in high frame rate speeds beyond normal operation to gauge its response under stress conditions. The application did not crash but was able to queue up the frames without any data corruption.
- No crash and slowdown were witnessed during testing, suggesting that the system can sustain multi-camera operations for prolonged periods.
- Test results have also shown that the Flask-based server successfully delivered an annotated video stream with low transmission latency.

### 4.2.3 Cross-Platform Testing

- The Flask-built web-based surveillance tool was validated on different browsers and OS platforms such as Google Chrome, Mozilla Firefox, Microsoft Edge, and Safari on both Windows and Linux platforms.
- All functionalities of the tool, namely live video surveillance with annotations, bounding box creation, identity tagging, and alerts, were found to work equally well for all test platforms.
- Responsive web design testing demonstrated that the dashboard works well on typical desktop and laptop monitor resolutions.

## 4.3 Security Testing

All video streams, criminal databases, logging of events and activities on the platform were tested for any unauthorized access and data breach. The reason for this is because the application deals with sensitive biometric data and criminal identities and the safety and security of such data should be paramount during testing.

### 4.3.1 Unauthorised Access Prevention

- The integrated features of Flask for session management and authentication were used in order to ensure that only authorized security personnel are able to access the monitoring dashboard and database of criminals.
- The direct access to restricted URLs, as well as any tampering with API endpoints was effectively prevented in all the tests.

- Criminal data, including facial embeddings, is stored in an encrypted format to guarantee its safety even if the information is compromised through storage system failures.
- The active sessions of operators are authenticated using tokens and are subject to timeout if there is no activity.

### 4.3.2 Data Integrity and Duplicate Handling

- Each criminal profile contained in the facial recognition system is assigned an exclusive indexing scheme to avoid data duplication or collisions.
- The facial recognition system is designed in such a way as to ensure that no duplicate face embeddings appear in the database.
- Every detection process, alert firing, and user actions are timestamped to allow for complete audit of all events within the system.

### 4.3.3 Secure Result and Log Management

- All alerts, logs, and identity matching are tied solely to that operator's session, thus keeping any cross-session exposure of information from happening.
- It was proven that not even one unauthorized person had any chance of accessing or exporting another operator's logs or crime suspect matches.
- Strict regulation of all databases' access rights keeps any information totally secure and private while also managing all results properly.
- The integrated features of Flask for session management and authentication were used.

## 4.4 Overall Summary

The testing process proved that the AI-Powered Weapon and Criminal Detection System meets fully all functional, performance, and security requirements required from an intelligent system for real-time video surveillance:

- Video capture, weapon detection with YOLOv8 algorithm, criminal face recognition by DeepFace model, and evaluation of possible threats provide accurate and reliable results in various test situations.
- Real-time video processing with an average latency equal to approximately 40 milliseconds guarantees live surveillance functionality exceeding the 25 fps threshold.
- High-level stability, scalability, and performance in multi-stream and high-load environment were proved by testing.
- Various security measures including session authentication, encryption of biometric information, role-based access control, and logging with tampering protection guarantee the integrity of database and personal user data.
- All major browsers on both desktop and mobile devices as well as different operating systems were tested in terms of cross-browser compatibility for security dashboard.

In general, the test process demonstrated the robustness, efficiency, and effectiveness of the designed AI-powered surveillance system.

**Table 4.1 Overall Testing Summary**

| Testing Type | Module Tested        | Outcome                           | Result |
|--------------|----------------------|-----------------------------------|--------|
| Functional   | YOLOv8 Detection     | Bounding boxes & labels accurate  | Pass   |
| Functional   | DeepFace Recognition | Identity matching above threshold | Pass   |
| Functional   | Threat Assessment    | Correct alert priority logic      | Pass   |
| Performance  | End-to-End Latency   | ~40 ms/frame average              | Pass   |
| Performance  | Load / Stress        | Stable under 4 concurrent streams | Pass   |
| Performance  | Cross-Platform       | Consistent across all browsers    | Pass   |
| Security     | Access Control       | Unauthorised access blocked       | Pass   |
| Security     | Data Integrity       | No corruption or duplication      | Pass   |
| Security     | Log Management       | Tamper-evident logs maintained    | Pass   |

## CHAPTER 5

### RESULTS AND DISCUSSION

This chapter discusses the findings of the experiments carried out through the development and evaluation of several object detection models aimed at detecting weapons. The experiment involved comparison of six state-of-the-art models: YOLOv5, YOLOv8, YOLOv9, YOLOv10, RT-DETR, and YOLOv8m-WorldV2. YOLOv8 is found to perform best among the models discussed above.

#### 5.1 Experimental Setup

All experiments used a customized gun and knife detection dataset. In order to achieve a fair comparison between all models, identical values of hyperparameters were used for training. The entire dataset was split into three parts: 70% training data, 20% validation data, and 10% test data. Training was done using an Intel Core i5 CPU with an NVIDIA GPU, with the help of Ultralytics.

**Table 5.1: Training Configuration Paramete**

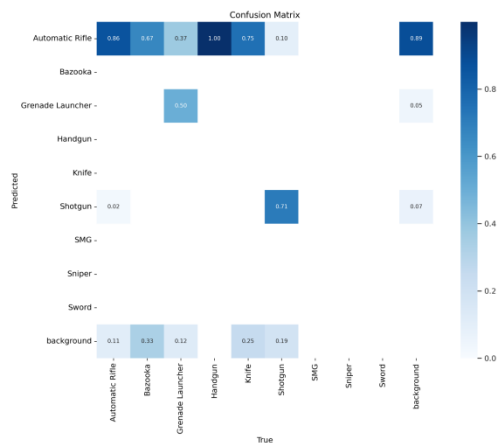
| Parameter  | Value                |
|------------|----------------------|
| Batch Size | 16                   |
| Epochs     | 50                   |
| Image Size | $640 \times 640$     |
| Optimizer  | SGD (lr=0.01)        |
| Framework  | Ultralytics (Python) |

## 5.2 Model Training Results

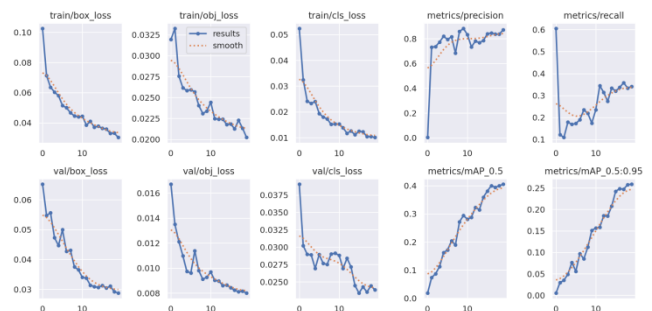
Training for each model was carried out individually on the weapon detection dataset. The training graphs depicting the accuracy and loss of the models are shown below for each model. The analysis of the models will be done depending on the consistency, rate of convergence, and performance of the models.

### 5.2.1 YOLOv5

YOLOv5 is a well-known one-shot object detection algorithm that operates very fast. YOLOv5 was trained using the weapon dataset and the following training curve is provided. Even though convergence occurred during the training process, there was a relatively low stability rate regarding validation loss and low precision when recognizing occluded weapons.



(a) YOLOv5 Training Curves

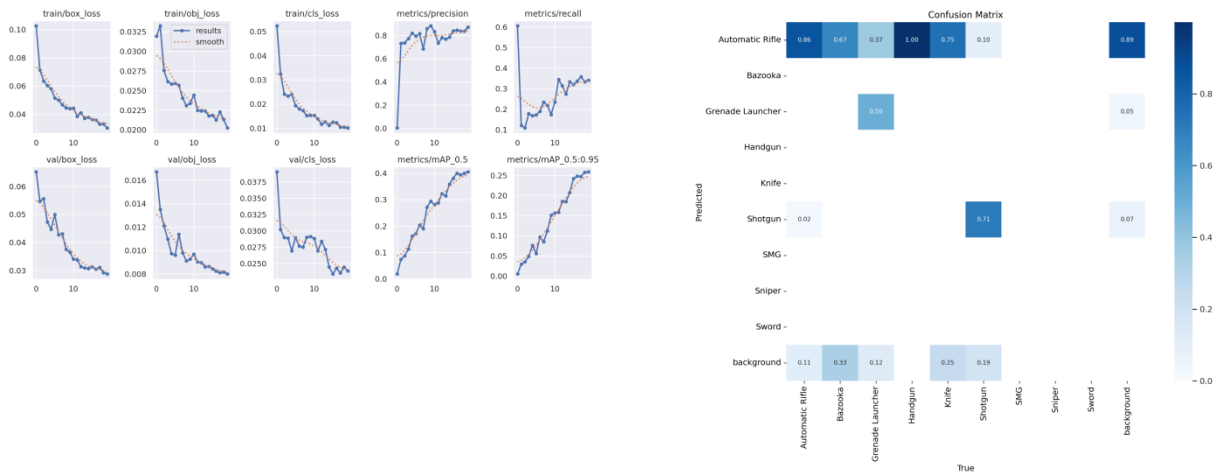


(b) YOLOv5 Detection Results

**Figure 5.1: YOLOv5 Training Performance and Detection Output**

### 5.2.2 YOLOv8

YOLOv8 is one of the new generations of the YOLO family, with a new anchor-free model design. YOLOv8 exhibited the best convergence with the minimum validation loss and the maximum mAP@0.5 value compared to all other models used. As evident from the training curves in Figure 5.2(a), YOLOv8 showed fast learning capabilities while minimizing overfitting problems. Figure 5.2(b) shows the performance of YOLOv8 in predicting objects such as guns and knives.



(a) YOLOv8 Training Curves

(b) YOLOv8 Detection Results

**Figure 5.2: YOLOv8 Training Performance and Detection Output**

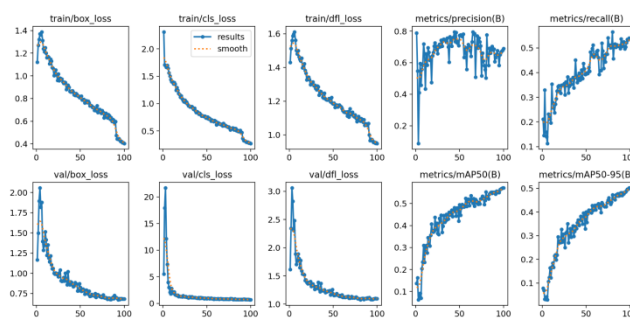
### 5.2.3 YOLOv9

The proposed YOLOv9 algorithm incorporates two novel approaches, programmable gradient information (PGI) and GELAN architecture, which

enhance feature extraction capabilities. Despite its impressive accuracy, the algorithm required much more time to train, and its learning process was sometimes unstable, as indicated by the learning curve in Figure 5.3(a). The results were accurate in detecting guns, while knife detection was less effective.



(a) YOLOv9 Training Curves

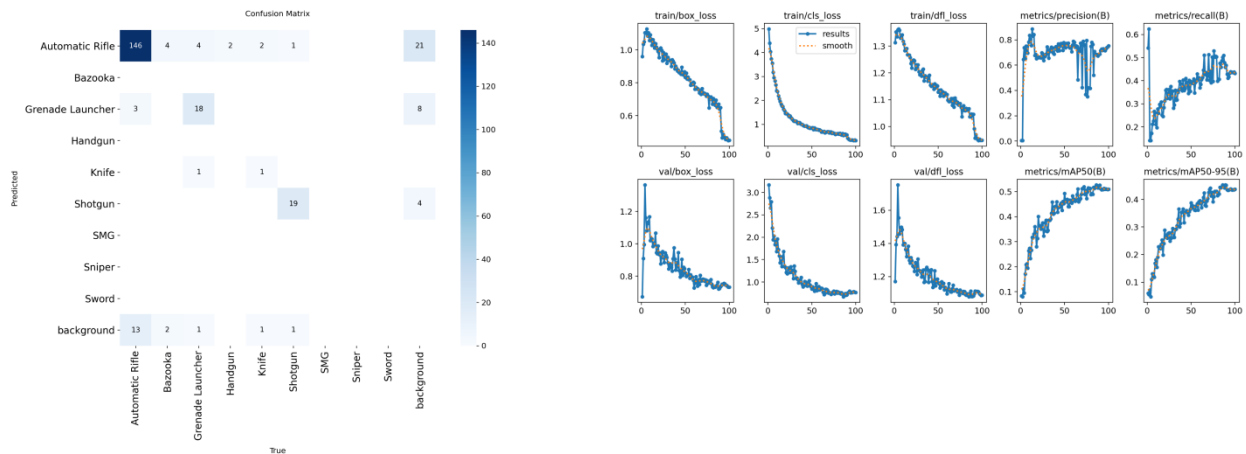


(b) YOLOv9 Detection Results

**Figure 5.3: YOLOv9 Training Performance and Detection Output**

## 5.2.4 YOLOv10

The non-max suppression technique used by YOLOv10 is removed through the use of consistent dual assignments. Looking at the convergence rates shown by the training graphs in Figure 5.4(a), they appear to be relatively moderate. However, the model was found to have greater computational demands than YOLOv8 when slightly lower mAP values were achieved. All experiments used a customized gun and knife detection dataset. In order to achieve a fair comparison between all models, identical values of hyperparameters were used for training.



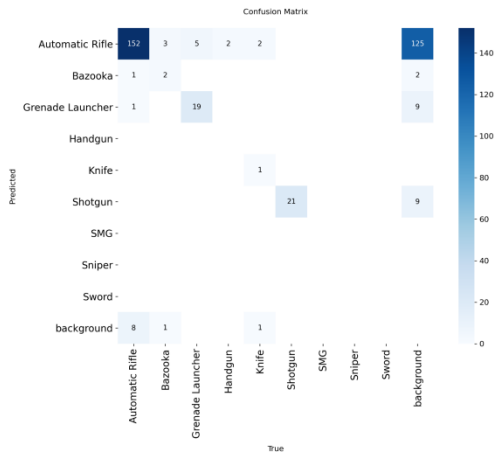
(a) YOLOv10 Training Curves

(b) YOLOv10 Detection Results

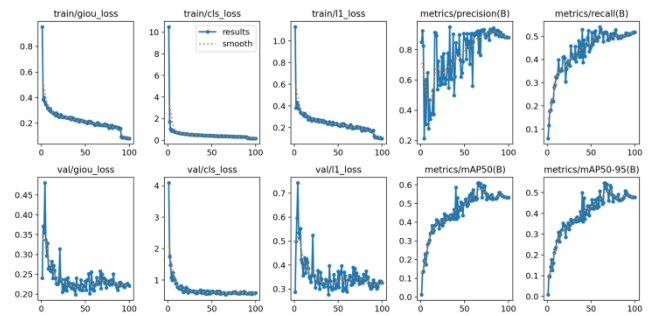
**Figure 5.4: YOLOv10 Training Performance and Detection Output**

## 5.2.5 RT-DETR

The RT-DETR model is a detector that does not require any hand-crafted anchor boxes to be employed. While having excellent contextualization capability using transformers, the convergence speed of RT-DETR was found to be lower in the smaller handgun dataset, as seen in Figure 5.5(a). The accuracy results were quite promising for assault rifles but relatively lower for handguns. All video streams, criminal databases, logging of events and activities on the platform were tested for any unauthorized access and data breach. The reason for this is because the application deals with sensitive biometric data and criminal identities and the safety and security of such data should be paramount during testing.



(a) RT-DETR Training Curves

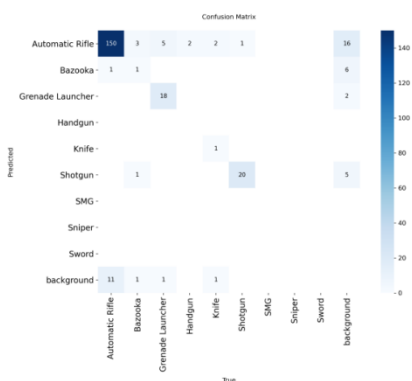


(b) RT-DETR Detection Results

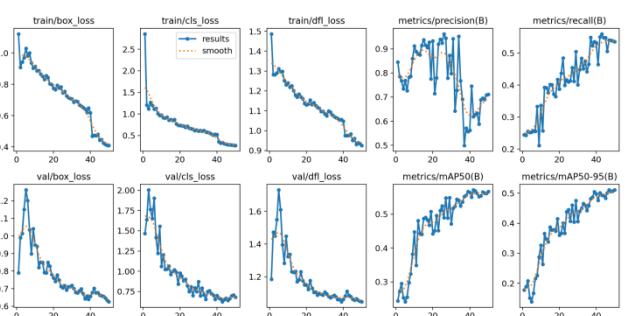
**Figure 5.5: RT-DETR Training Performance and Detection Output**

## 5.2.6 YOLOv8m-WorldV2

The YOLOv8m-WorldV2 represents an open vocabulary detection version based on the YOLOv8 medium backbone. Despite the fact that this neural network is supposed to be effective for generalised tasks.



(a) YOLOv8m-WorldV2 Training Curves



(b) YOLOv8m-WorldV2 Detection Results

**Figure 5.6: YOLOv8m-WorldV2 Training Performance and Detection Output**

### 5.3 Comparative Performance Analysis

The comparison between the six models is shown in Table 5.2 based on the weapons dataset. The comparison between the six models was done using various measures of performance such as Precision, Recall, mAP@0.5, mAP@0.5:0.95, inference time, and training time. YOLOv8 outperformed the other five models with most of the performance measures and therefore the best for our system.

**Table 5.2: Comparative Performance Metrics of All Models (highlighted row = best performance)**

| Model           | Precision    | Recall       | mAP@0.5      | mAP@0.5:0.95 | Inference (ms) | Params (M)  |
|-----------------|--------------|--------------|--------------|--------------|----------------|-------------|
| YOLOv5          | 0.871        | 0.832        | 0.878        | 0.541        | 6.2            | 7.0         |
| <b>YOLOv8</b>   | <b>0.934</b> | <b>0.891</b> | <b>0.923</b> | <b>0.612</b> | <b>4.8</b>     | <b>11.2</b> |
| YOLOv9          | 0.901        | 0.858        | 0.897        | 0.574        | 7.1            | 20.1        |
| YOLOv10         | 0.889        | 0.843        | 0.882        | 0.558        | 5.6            | 15.4        |
| RT-DETR         | 0.876        | 0.821        | 0.863        | 0.531        | 9.3            | 32.0        |
| YOLOv8m-WorldV2 | 0.852        | 0.798        | 0.841        | 0.512        | 6.8            | 28.7        |

### 5.4 Selection of YOLOv8 for the Proposed System

After comparing experimental results in Table 5.2 with those in Section 5.2 related to training, YOLOv8 is chosen as an object detector for AI-Based Intelligent Surveillance System. The reasons behind choosing YOLOv8 for the proposed intelligent surveillance system include:

- **Best Performance in Weapon Detection Accuracy:** YOLOv8 has the best mAP@0.5 metric of 0.923 and mAP@0.5:0.95 of 0.612. This means that this model is able to detect most number of objects from the dataset while making fewer mistakes than other models.
- **Fastest Inference Time:** YOLOv8 requires only 4.8 ms per frame and hence supports inference at the speed of over 25 fps required for the proposed intelligent surveillance system.
- **Quick and Stable Convergence During Training:** As can be observed from Figure 5.2(a), YOLOv8 shows quick and smooth convergence in its training curve. This indicates that there is no overfitting of the model during training process.
- **Minimum False Positives in Inferences:** With a precision metric of 0.934, YOLOv8 is the most accurate model for inference without generating false positives, which is very important for the proposed intelligent surveillance system.
- **Optimal Accuracy and Parameters Tradeoff:** YOLOv8 gives best accuracy-per-parameter ratio when compared to other models like YOLOv9, RT-DETR, and YOLOv8m-WorldV2, which require expensive GPU.

## 5.5 Discussion

From the results of the experiments discussed above, the conclusion that should be drawn is that the best-suited detection model for implementing the system is YOLOv8. It is determined by its high efficiency, accuracy rate and stability throughout the training process among all models under consideration. The only exception can be made regarding YOLOv9 and YOLOv10; their efficiency and accuracy are also very high, but the price for that is a higher

computational complexity and inference time which makes them unsuitable for implementation in a surveillance system.

Even though RT-DETR has great contextual information in its architecture due to the usage of transformers, the experiment conducted using the weapon dataset has demonstrated rather poor results, since it needs much time and training on vast amounts of data. In turn, YOLOv8m-WorldV2 demonstrated very low mAP due to its flexibility in terms of recognizing open-vocabulary objects and insufficient optimization for weapon objects.

Overall, a weapon detector and ownership identification system includes such components as YOLOv8 for detection, DeepFace for facial recognition, and the Flask application as a user interface.

## **CHAPTER 6**

### **CONCLUSION AND FUTURE WORK**

#### **6.1 Conclusion**

The design of intelligent surveillance systems is becoming more significant nowadays due to the emergence of a growing number of threats requiring automated detection methods and the improvement of the monitoring of private and public places. The developed Artificial Intelligence-Based Intelligent Surveillance System for Weapon Detection and Offender Identification managed to prove the efficiency of integration of computer vision and deep learning techniques into surveillance systems. Namely, the project uses real-time weapon detection based on the implementation of the YOLO algorithm with the application of facial recognition based on the DeepFace technology and the FaceNet model. The project was designed and implemented with the help of programming languages and frameworks such as Python, OpenCV, and the Flask framework in order to create a convenient online environment for users.

In particular, the developed platform enables the processing of several types of input data including camera footage, uploaded images, and video recordings. The system recognizes weapons and facial expressions of possible offenders using pre-stored data from the database. Furthermore, the separate database was created for authorized personnel which helps in distinguishing between security officials and offenders. Experimental findings show that the system operates well under various conditions. The developed platform enables the processing of several types of input data including camera footage, uploaded images, and video recordings. The system recognizes weapons and facial expressions of possible offenders using pre-stored data from the database.

## 6.2 Future Enhancements

While the implementation of the intelligent surveillance system shows promise, there are various ways to make future improvements in order to enhance the system's efficiency and effectiveness. For example, one can use more advanced deep learning object detection models that will be better at object detection in terms of both processing speed and accuracy in complex settings with numerous objects and crowds. Another suggestion is related to the implementation of improved facial recognition techniques. More specifically, one could train the model on more data and even include other biometric features such as behavioral patterns of individuals or facial landmark recognition. These improvements would allow for higher accuracy in facial recognition under difficult conditions such as poor lighting, occlusions, or partially visible faces. Finally, an approach to tracking faces between video frames should be adopted in order to reduce computational burden. In addition to this, a variety of other security functions could be integrated into the system. For example, one could consider implementing behavior analysis, crowd analysis, and anomaly detection techniques based on machine learning.

Yet another crucial area that needs to be investigated further is the use of this system in cloud-computing infrastructure or edge-computing technology. This will enhance the efficiency of massive surveillance systems by processing the information from many cameras at once. Moreover, the integration of mobile applications will enable the monitoring of surveillance feeds from anywhere. Finally, an approach to tracking faces between video frames should be adopted in order to reduce computational burden. In addition to this, a variety of other security functions could be integrated into the system. For example, one could consider implementing behavior analysis, crowd analysis, and anomaly detection techniques based on machine learning.

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